

VOLODYMYR TKACHUK

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EDUCATION

- Now **PhD in Computing Science** (advisors: Csaba Szepesvári & Xiaoqi Tan) – **University of Alberta**
- 2023 **MSc in Computing Science** (advisor: Csaba Szepesvári) – **University of Alberta**
- 2021 **BSc in Electrical Engineering – University of Waterloo**
- **91.9% GPA** in final two years, and academically **ranked 2nd** in class of 34 students for two terms

PUBLICATIONS

- [1] **Trajectory Data Suffices for Statistically Efficient Learning in Offline RL with qpi-Realizability and Concentrability**
Volodymyr Tkachuk, Gellért Weisz, Csaba Szepesvári (NeurIPS 2024)
- [2] **Regret Minimization via Saddle Point Optimization**
Johannes Kirschner, Alireza Bakhtiari, Kushagra Chandak, Volodymyr Tkachuk, Csaba Szepesvári (NeurIPS 2023)
- [3] **On Efficient Planning in Large Action Spaces with Applications to Cooperative Multi-Agent Reinforcement Learning**
Volodymyr Tkachuk (MSc Thesis)
- [4] **Efficient Planning in Combinatorial Action Spaces with Applications to Cooperative Multi-Agent Reinforcement Learning**
Volodymyr Tkachuk, Seyed Alireza Bakhtiari, Johannes Kirschner, Matej Jusup, Ilija Bogunovic, Csaba Szepesvári (AISTATS 2023)
- [5] **Investigating action encodings in recurrent neural networks in reinforcement learning**
Matthew Kyle Schlegel, Volodymyr Tkachuk, Adam M. White, Martha White (TMLR 2023)

PUBLIC TALKS

- 2024 **Trajectory Data Suffices for Statistically Efficient Learning in Offline RL with qpi-Realizability and Concentrability** (Amii AI Seminar 2024) | [Video](#)

AWARDS AND SCHOLARSHIPS

- 2024 Alberta Graduate Excellence Scholarship (**\$12,000**)
- 2023 Graduate Teaching Award
- 2021 NSERC - Canada Graduate Scholarship – Master’s (**\$17,500**)
- 2021 Walter H Johns Graduate Fellowship (**\$5800**)
- 2020 Gerry Heckman Scholarship (**\$3000**)
- 2018 Dan and Anik Colquhoun Award (**\$2000**)
- 2015 University of Waterloo President’s Scholarship (**\$2000**)

RESEARCH EXPERIENCE

- 2021 **Research Assistant**
University of Alberta – Advised by Martha White

- 2020 **Research Assistant**
University of Alberta – Advised by Matthew E. Taylor
- 2019 **Research Engineering Intern**
Apple – Acoustics Special Projects Group

PROFESSIONAL EXPERIENCE

- 2019 **Audio System Tuning Intern**
Apple – Telephony Tuning Team
- 2018 **Advanced CMOS Sensor Development Intern**
Teledyne DALSA – CMOS Sensor Products Team
- 2017 **Digital Tool Developer and Verification Engineering Intern**
Peraso Technologies – Digital Verification Team
- 2016 **Web Developer**
University of Waterloo – Undergraduate Recruitment Group
- 2016 **Software Engineering Intern**
T. Hong & Co. – Web Development Team

TEACHING EXPERIENCE

- 2024 **Instructor** (Fall term)
University of Alberta – CMPUT 267 (Machine Learning I) [[website](#)]
- 2024 **Graduate Teaching Assistant** (Winter term)
University of Alberta – CMPUT 267 (Basics of Machine Learning) [[website](#)]
- 2023 **Graduate Teaching Assistant**
University of Alberta – CMPUT 605 (Theoretical Foundations of Reinforcement Learning) [[website](#)]
- 2022 **Graduate Teaching Assistant**
University of Alberta – CMPUT 653 (Theoretical Foundations of Reinforcement Learning) [[website](#)]
- 2021 **Graduate Teaching Assistant**
University of Alberta – CMPUT 396 (Intermediate Machine Learning) [[website](#)]

SKILLS

Languages and Tools

Python, C/C++, MATLAB, Vim, Git